FFVP

(Fédération Française de Vol en Planeur)

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SLM Certification Programme

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0.1 Amendment Record

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01	Initial Issue	

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1. Introduction

This Certification Programme (CP) defines the compliance approach for an EASA Supplemental Type Certificate (STC) allowing extension of the operational life limit of selected gliders (Schempp-Hirth and LTB Lindner) using a Structural Life Monitoring (SLM) system. By tracking the Equivalent Fatigue Index (EFI) of each individual glider, the SLM system may support operation beyond the 12,000-hour safe-life limit, provided the aircraft remains within the approved SLM applicability domain and complies with the approved CONOPS, Instructions for Continued Airworthiness (ICA), Installation, Operating and Maintenance Manual (IOMM), required inspections, and EFI go/no-go criteria.

Some glider models included in the STC scope may have been originally certified by national European aviation authorities or under JAA arrangements before EASA assumed responsibility for the product. For the purpose of this STC, only aircraft types with an EASA-recognized Type Certificate Data Sheet and listed in the EASA product list are considered.

The Certification Programme will be updated, as required, to reflect any relevant changes to the project.

1.1 Project Description

The STC introduces a system-based life-extension method using an accelerometer-based recorder installed in the fuselage, in a rigid area close to the C.G. and wing spar, and an analytical EFI calculation. No structural modification is required. The intent of this STC is not to change the structural fatigue requirements or limits defined in the original Type Certificate (TC), but to demonstrate that individual-aircraft usage monitoring, using SLM, provides an equivalent level of safety compared to the conservative assumptions used in the original CS-22 fatigue substantiation. In a nutshell, SLM calculates the glider's cumulative EFI at the measured usage rate and compares it with the certification reference EFI rate derived from the Kossira & Reinke Utility reference spectrum. The method does not claim to measure the absolute physical damage state of the composite structure.

The following items shall be approved within the STC:

- SLM methodology (principles, assumptions, limitations, EFI calculation method and selected reference spectrum)
- SLM hardware installation (as minor change via CS-STAN)
- SLM operational and maintenance procedures (Concept of Operations / CONOPS), including continued-airworthiness processes linked to EFI status and decision criteria for operation up to the certification EFI limit

1.2 Project History

Glider types involved in the experimentation, under the STC, have accumulated extensive operational experience. The STC builds upon existing fatigue-life knowledge and introduces monitoring-based operation up to the certification EFI limit rather than structural margin re-evaluation or structural redesign.

1.3 Design Description

The SLM modification adds a lightweight accelerometer-based recorder, post-flight analysis tools, and updated AFM/AMM/AMP instructions where applicable. No aerodynamic or structural modifications are required, and the certified aircraft flight envelope is not changed. However, SLM credit is limited to operation within the approved SLM applicability domain, namely Utility-category / non-aerobatic operation as defined in the SLM Methodology Report, CRI, ICA and IOMM.

Hardware: An SLM recorder installed in the fuselage, in a rigid area close to the C.G. and wing spar, to measure in-flight accelerations used for EFI calculation.

Software/Process: A data analysis toolset for post-flight EFI evaluation based on the Palmgren-Miner rule, the Kossira & Reinke counting method and the associated Utility reference spectrum.

Key Advantage: this approach does not require reopening original Type Certificate (TC) structural justification documents, as it relies on the certification EFI limit corresponding to the safe-life limit (typically 12,000 hours).

1.4 Design Organisation Resources

FFVP DOA provides engineering, airworthiness and programme management resources. Flight dynamics as well as structure and maintenance experts support the development of the SLM methodology.

1.5 Design Subcontractors

Two companies are going to support this programme:
- Schempp-Hirth Flugzeugbau GmbH
- LTB Lindner

They are the TC holders for the gliders part of this STC.
They will review the STC documents and contribute with their expertise, to the establishment of recurring test covering both composite and non-composite parts. Since they are the TC holders and have access to the complete STC data package, they will be free to develop modifications and associated Service Bulletins to distribute SLM more broadly.

1.6 Technologies and Process Description

Technologies include lightweight sensing, digital data recording, analytical EFI modelling, and associated data processing. Processes include calibration, data validation, cumulative EFI tracking and remaining EFI potential calculation, the latter providing the opportunity for operation beyond 12,000 hours up to the certification EFI limit.

Although innovative in the civil glider context, the SLM solution relies on estimation methods (e.g. Palmgren-Miner rule) that are well established and accepted across the aerospace industry. Similar measured-usage principles have been used in aviation usage monitoring and Individual Aircraft Tracking, particularly in military aviation. The SLM proposal does not attempt to reproduce those military approaches directly or claim that they are directly transferable to gliders. The relevant precedent is the general principle that measured usage can be compared with a conservative reference usage assumption, provided the method is controlled, conservative, and supported by inspections. SLM leverages the significant work of Kossira & Reinke, who established filtering, cycle-counting methods and used Markov transition matrices to develop glider load spectra officially referenced by EASA in Certification Memorandum CM-S-006.

2. Type Certification Basis and Environmental Protection Requirements

The STC certification basis follows CS-22, with primary focus on CS 22.627 Fatigue. Even if no environmental requirements apply to this STC, it contributes to lower the environmental impact by increasing life usage of composite gliders, for which no recycling solutions are available.

2.1 Type Certification Basis

Applicable requirements: CS-22, Part 21 Subpart E. Primary requirement: CS 22.627 Fatigue.
Since there are no Interpretative Material (IM) or Acceptable Means of Compliance (AMC) to support the proposed SLM solution, a CRI will have to be issued to provide the required Means of Compliance material for the SLM fatigue-monitoring method.
The CRI shall cover acceptable:
- EFI calculation method based on the Palmgren-Miner rule
- Kossira & Reinke Utility reference spectrum
- conservative composite severity parameter
- onboard recording solution and installation/recurring calibrations
- post-processing solution and tool validation
- treatment of invalid, missing and non-flight data
- CONOPS and Instructions for Continued Airworthiness (including impact on AFM/AMM/AMP where applicable)
- limits of proposed methodology and SLM applicability, including exclusion of aerobatic and non-standard operations
- go/no-go criteria based on EFI total < 1/3 and projected EFI rate

This CRI should provide means of compliance allowing a glider equipped with SLM to operate beyond the originally established safe-life limit, provided that the cumulative EFI remains below the certification EFI limit and the approved inspection and continued-airworthiness provisions are satisfied.

2.2 Equivalent Safety Findings (ESF), Deviations

None anticipated. No deviations from CS-22 are required.
- The STC stays within CS-22 intent
- Equivalence is shown analytically and operationally
- Any ESF discussion, if needed, will be embedded in the CRI

2.3 Special Conditions

None expected.

2.4 Environmental Protection

Not applicable to this STC.

2.5 Operation Suitability Data (OSD) – Minimum Equipment List (MEL)

No OSD items are required since there is no change to the operation of the glider within the approved flight envelope. MEL is not affected before the TC safe-life limit is reached.
The SLM recorder is optional prior to reaching the original TC safe-life limit (e.g. 12,000 hours). However, to benefit from SLM, the recorder needs to be operated before the safe-life limit so that valid measured-usage data can be credited.
Once operation exceeds the TC safe-life limit, the SLM system becomes mandatory, and its availability, calibration status, validated EFI reports and associated inspections become conditions for continued airworthiness as defined in the approved ICA and IOMM.

3. Certification Process

3.1 Means of Compliance

MoC shall be the result of the proposed CRI.

- EFI calculation methods: fatigue-equivalence demonstration using the Palmgren-Miner rule as a common comparison framework

- reference load spectrum and composite severity parameter: calibration of the EFI model against the accepted certification reference exposure, using the Kossira & Reinke Utility spectrum and the conservative composite parameter defined in the SLM Methodology Report. This is required to calculate cumulative EFI for measured operation and non-instrumented substitution

- onboard recording solution: sensor installation and calibration procedures

- post processing solution: toolset validation method

- CONOPS and Instructions for Continued Airworthiness: AFM/AMM/AMP integration where applicable, mandatory inspections, EFI status reports, and Go / No-Go criteria

- limits of SLM applicability/methodology (e.g. approved operating domain, exclusion of aerobatic or non-standard operations, and substitution procedure when SLM data is inaccurate or unavailable)

No flight testing is required since data will be gathered transparently during normal operations.

Instructions for Continued Airworthiness (ICA) required to support SLM shall be prepared and provided in accordance with Certification Specification CS-22.1529.

It should be understood that SLM is mandatory for operation beyond the originally certified safe-life limit and that sensor calibrations, validated EFI status reports and inspections are release-to-service conditions.

While the glider is operated before the originally certified safe-life limit, no measured-usage credit can be granted for periods not covered by valid SLM data. These periods are substituted at the reference EFI rate in accordance with the approved ICA and data-processing procedure.

Abnormal and failure cases shall be addressed within the ICA and CRI, including loss of recorded data, missed calibration, sensor malfunction, or data inconsistency. In such cases, continued operation beyond the certified safe-life limit is not permitted unless corrective actions are taken and compliance with the approved SLM process is restored.

3.2 Certification Schedule

The schedule should be provided following the phases described above and should include the major milestones, for example:

T0: CP Submission.

T0 + 3 Months: CRI Acceptance.

T0 + 12 Months: Full compliance documentation submitted.

T0 + 18 Months: Anticipated STC issuance.

3.3 Approval of Flight Conditions

No special flight conditions required. No certification flight testing.

3.4 Engine/Propeller Certification and Interactions with Aircraft

Not applicable; SLM does not affect propulsion.

3.5 Equipment Qualification

Recorder is qualified and validated through analytical justification using MEMS sensor manufacturer specifications as well as functional tests, calibration checks, and data-integrity verification. This includes a calibration with 6-face rotation using gravity as a reference as well as an installation calibration, with glider in flight level attitude.

3.6 Control of Production and Configuration

Recorder installation configuration is tracked using Certification Specifications for Standard Changes and Standard Repairs (CS-STAN) process. Each aircraft installation is traceable.

- One recorder per aircraft
- Aircraft-specific serialisation of data (data specific to glider)
- Central repository as a controlled source of truth
- Protection against selective data deletion or filtering (aircraft logbook is the reference)

The central SLM data repository constitutes the authoritative source for EFI data used for airworthiness decisions. EFI status reports used for ARC and continued airworthiness shall be generated and validated in accordance with the approved SLM procedures.

3.7 Compliance Documentation

Compliance includes:
- SLM Certification Programme

- Certification Review Item (CRI) or equivalent means-of-compliance agreement
- SLM Methodology Report
- Instructions for Continued Airworthiness (ICA)
- Installation, Operating and Maintenance Manual (IOMM)
- SLM data-processing tool validation report
- SLM data repository / data integrity procedure
- Master Compliance Report

3.8 Type Design Definition

The change does not affect the physical configuration of the aircraft nor the certified structural fatigue limit defined in the Type Certificate.
The approved change introduces an alternative, monitoring-based means to demonstrate when the certification EFI limit is reached, on an individual-aircraft basis.

- No impact on form, fit and function; the aircraft configuration remains identical.
- No change to certified operating limitations.
- No new mandatory equipment within the originally certified life limit.
- SLM equipment, calibration, validated EFI reports and inspections become conditions for continued airworthiness when operating beyond the safe-life limit.

3.9 Master Document List

An MDL will be issued listing CP, CRI, SLM Methodology Report, ICA, IOMM, validation reports, data repository / data integrity procedure, approved calculation package and compliance data.

3.10 Final Declaration of Compliance

Issued upon completion and EASA acceptance of compliance data / reports.

3.11 Proposed EASA Involvement

Compliance Data / Activity	Proposed EASA Involvement	Rationale
CRI (SLM Means of Compliance)	Approval	To validate the overall SLM concept and operation up to the certification EFI limit. Agree on the MoC statement.
SLM Methodology Report	Review	To agree with the mathematical approach to EFI calculation and certification EFI limit, including agreement on selected Utility reference spectrum and composite severity parameter.
Instructions for Continued Airworthiness	Review	To agree on compliance with Part ML and continued-airworthiness provisions, including inspections, release-to-service conditions, and EFI status reporting.
Installation, Operating & Maintenance	Review	Standard procedures following CS-STAN principles and associated AFM/AMM/AMP where applicable, including installation calibration, recorder operation, maintenance and data handling.
Master Compliance Report	Approval	Final sign-off on the demonstration of compliance for the STC.

4. List of Annexes

Annex A: CONOPS

Annex B: Master Document List

Annex C: Compliance Check List

ANNEX A: CONOPS

CONOPS can be divided in 3 main phases:

Phase 1 — Instrumentation and Data Collection (below TC safe-life limit)
This phase covers the period during which the SLM recorder is installed and the glider has not yet reached its safe-life limit. It includes:

- Sensor installation in accordance with CS-STAN / Form 123
- Sensor installed in the fuselage, in a rigid area close to the C.G. and wing spar, and calibrated with the glider in flight-level attitude and wings level
- Annual 6-face calibration during the annual inspection
- Compliance with Part ML; installation and annual calibration results recorded in the aircraft logbook
- Missing data policy: if a flight occurs without valid SLM data (sensor fault, data corruption, missed calibration) or includes a phase excluded from accelerometer-based processing, the EFI for that flight or excluded phase is substituted at the Kossira & Reinke Utility reference EFI rate for the corresponding duration or operational exposure, as defined in the approved ICA and data-processing procedure
- Flight data uploaded to the controlled central SLM repository; post-processing tools providing EFI status reports

Phase 2 — EFI Evaluation when reaching TC safe-life limit
This phase includes:

- a dedicated fatigue inspection of the aircraft condition (equivalent in scope to the existing 1,000-hour inspections).
- an assessment of cumulative EFI total, including measured EFI and reference-substituted EFI for non-instrumented, invalid-data or excluded non-flight periods, to determine whether operation beyond the TC safe-life limit may be permitted up to the certification EFI limit EFI = 1/3.
- inspection beyond the composite structure, as well as an overall assessment of glider’s condition (incidents/repairs).

Phase 3 — Operation Beyond TC safe-life limit
This phase is a continuation of Phase 1 monitoring operations, with the following additional provisions:

- Go / No-Go decision at each Airworthiness Review Certificate (ARC) renewal or other approved review point, based on the validated SLM EFI status report
- Go criterion: cumulative EFI total < 1/3, AND the projected EFI rate confirms that EFI = 1/3 will not be reached before the next scheduled inspection or approved review point
- Recurring structural inspection continuing from Phase 2, with a maximum interval of 1,000 hours or 5 years, whichever is sooner.
- This process continues until the aircraft has exhausted its remaining EFI potential, another approved continued-airworthiness limit is reached, or the aircraft is retired from service

ANNEX B: Master Document List

Reference	Title	Issue	Date
SLM-CP	Certification Programme		
SLM-CRI	Certification Review Item		
SLM-MR	Methodology Report		
SLM-ICA	Instructions for Continued Airworthiness		
SLM-IOMM	Installation, Operating and Maintenance Manual		
SLM-MCR	Master Compliance Report		

ANNEX C: Compliance Check List

Req	Requirement Summary	MoC	Compliance Document	EASA Involvement	Status
CS 22.627	Alternative MoC for SLM EFI-based fatigue life management	CRI	SLM-CRI, SLM-MR	Approval	Open
CS 22.1529	Instructions for Continued Airworthiness, including inspections, release-to-service conditions and EFI status reporting	Doc. Review	SLM-ICA	Review	Open
CS 22.1585	Operating procedures for SLM recorder, installation calibration, annual 6-face calibration and data upload	Doc. Review	SLM-IOMM	Review	Open
Part ML	Continued airworthiness, Part ML process and ARC support using validated EFI status reports	Doc. Review	SLM-ICA, SLM-IOMM	Review	Open